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Digital Communication Systems

GATE ECE Communications · Previous Year Questions with Answer Key

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QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	28 questions
Marks 2	39 questions
Marks 8	1 question
Marks 10	2 questions
Answer key	at the end

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Marks 1 28 questions

Q1. GATE ECE 2026

Consider a discrete memoryless source with an alphabet of four source symbols. $s(t)$ is a multi-level ($-1, 0, +1, +2$) signal representing a long sequence of random symbols from the above source which is generating 10^4 symbols per second. Which of the following options is the correct value of equivalent Nyquist bandwidth of $s(t)$?

(A) 10 kHz (B) 64 kHz (C) 5 kHz (D) 20 kHz

Open solution

Q2. GATE ECE 2026

A wireless digital transmission scheme is using 16-QAM over an additive white Gaussian noise channel and a maximum-likelihood receiver. Consider the information bit rate from source to be 4×10^6 bits per second.

The minimum transmission bandwidth (in MHz) of the modulated signal necessary for optimum recovery of information at the receiver is ____ .

(rounded off to two decimal places)

Open solution

Q3. GATE ECE 2024

A digital communication system transmits through a noiseless bandlimited channel $[-W, W]$. The received signal $z(t)$ at the output of the receiving filter is given by $z(t) = \sum_n b[n]x(t - nT)$ where $b[n]$ are the symbols and $x(t)$ is the overall system response to a single symbol. The received signal is sampled at $t = mT$. The Fourier transform of $x(t)$ is $X(f)$. The Nyquist condition that $X(f)$ must satisfy for zero intersymbol interference at the receiver is _____.

(A) $\sum_{m=-\infty}^{\infty} X\left(f + \frac{m}{T}\right) = T$ (B) $\sum_{m=-\infty}^{\infty} X\left(f + \frac{m}{T}\right) = \frac{1}{T}$

(C) $\sum_{m=-\infty}^{\infty} X(f + mT) = T$ (D) $\sum_{m=-\infty}^{\infty} X(f + mT) = \frac{1}{T}$

Open solution

Q4. GATE ECE 2022

Let H(X) denote the entropy of a discrete random variable X taking K possible distinct real values. Which of the following statements is/are necessarily true?

(A) $H(X) \leq \log_2 K$ bits (B) $H(X) \leq H(2X)$

(C) $H(X) \leq H(X^2)$ (D) $H(X) \leq H(2^X)$

Open solution

Q5. GATE ECE 2022

A symbol stream contains alternate QPSK and 16-QAM symbols. If symbols from this stream are transmitted at the rate of 1 mega-symbols per second, the raw (uncoded) data rate is _____ mega-bits per second (rounded off to one decimal place).

Open solution

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Q6. GATE ECE 2021

A message signal having peak-to-peak value of 2 V , root mean square value of 0.1 V and bandwidth of 5 kHz is sampled and fed to a pulse code modulation (PCM) system that uses a uniform quantizer. The PCM output is transmitted over a channel that can support a maximum transmission rate of 50 kbps . Assuming that the quantization error is uniformly distributed, the maximum signal to quantization

noise ratio that can be obtained by the PCM system (rounded off to two decimal places) is ____ .

Open solution

Q7. GATE ECE 2021

A 4 kHz sinusoidal message signal having amplitude 4 V is fed to a delta modulator (DM) operating at a sampling rate of 32 kHz . The minimum step size required to avoid slope overload noise in the DM (rounded off to two decimal places) is ____ V.

Open solution

Q8. GATE ECE 2020

A digital communication system transmits a block of N bits. The probability of error in decoding a bit is α. The error event of each bit is independent of the error events of the other bits. The received block is declared erroneous of at least one of its bits is decoded wrongly. The probability that the received block is erroneous, is

(A) $1-(1-\alpha)^N$ (B) α^N (C) $N(1-\alpha)$ (D) $1-\alpha^N$

Open solution

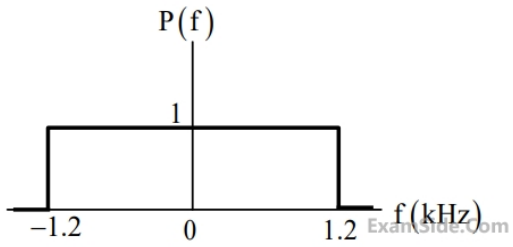
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Q9. GATE ECE 2017 Set 1

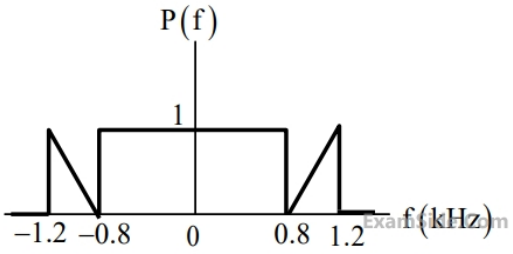
In a digital communication system, the overall pulse shape p(t) at the receiver before the sampler has the Fourier transform P(f). If the symbols are transmitted at the rate of 2000 symbols per second, for which of the following cases is inter symbol interference zero?

Open solution

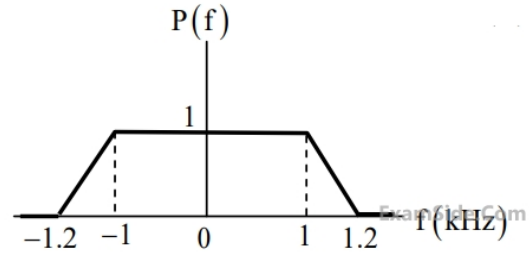
(A)



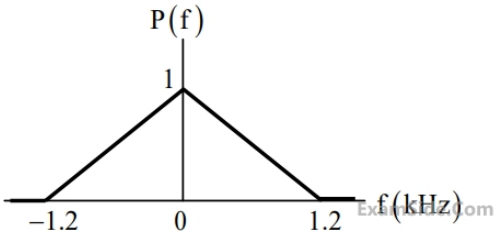
(B)



(C)



(D)



Q10. GATE ECE 2017 Set 1

Which one of the following statements about differential pulse code modulation (DPCM) is true?

Open solution

(A) The sum of message signal sample with its prediction is quantized

(B) The message signal sample is directly quantized, and its prediction is not used

(C) The difference of message signal sample and a random signal is quantized

(D) The difference of message signal sample with its predictions is quantized

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Q11. GATE ECE 2016 Set 2

A speech signal is sampled at 8 kHz and encoded into PCM format using 8 bits/sample. The PCM data is transmitted through a baseband channel via 4-level PAM. The minimum bandwidth (in kHz) required for transmission is _____.

Open solution

Q12. GATE ECE 2016 Set 1

Consider binary data transmission at a rate of 56 kbps using baseband binary pulse amplitude modulation (PAM) that is designed to have a raised-cosine spectrum. The transmission bandwidth (in kHz) required for a roll-off factor of 0.25 is _____.

Open solution

Q13. GATE ECE 2015 Set 3

The modulation scheme commonly used for transmission from GSM mobile terminals is

(A) 4-QAM
(B) 16-PSK
(C) Walsh-Hadamard orthogonal codes
(D) Gaussian Minimum Shift Keying (GMSK)

Open solution

Q14. GATE ECE 2014 Set 1

In a code - division multiple access (CDMA) system with N = 8 chips, the maximum number of users who can be assigned mutually orthogonal signature sequences is _____.

Open solution

Q15. GATE ECE 2013

The bit rate of a digital communication system is R kbits/s. The modulation used is 32-QAM. The minimum bandwidth required for ISI free transmission is

(A) R/10 Hz
(B) R/10 kHz
(C) R/5 Hz
(D) R/5 kHz

Open solution

Q16. GATE ECE 2004

In a PCM system, if the code word length is increased from 6 to 8 bits, the signal to quantization noise ratio improves by the factor

(A) 8/6
(B) 12
(C) 16
(D) 8

Open solution

Q17. GATE ECE 2004

In the output of a DM speech encoder, the consecutive pulses are of opposite polarity during time interval $t_1 \leq t \leq t_2$. This indicates that during this interval

(A) the input to the modulator is essentially constant.
(B) the modulator is going through slope overload.
(C) the accumulator is in saturation.
(D) the speech signal is being sampled at the Nyquist rate.

Open solution

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Q18. GATE ECE 2002

For a bit-rate of 8 kbps, the best possible values of the transmitted frequencies in a coherent binary **FSK** system are

(A) 16 KHz and 20 KHz
(B) 20 KHz and 32 KHz
(C) 20 KHz and 40 KHz
(D) 32 KHz and 40 KHz

Open solution

Q19. GATE ECE 2001

A band limited signal is sampled at the Nyquist rate. The signal can be recovered by passing the samples through

(A) an RC filter
(B) an envelope detector
(C) a PLL
(D) an ideal low-pass filter with the appropriate bandwidth

Open solution

Q20. GATE ECE 1998

In a PCM system with uniform quantization, increasing the number of bits from 8 to 9 will reduce the quantization noise power by a factor of

(A) 9 (B) 8 (C) 4 (D) 2

Open solution

Q21. GATE ECE 1998

Quadrature multiplexing is

(A) the same as FDM
(B) the same as TDM
(C) a combination of FDM and TDM
(D) quite different from FDM and TDM

Open solution

Q22. GATE ECE 1998

Compression in PCM refers to relative compression of

(A) higher signal amplitudes
(B) lower signal amplitudes
(C) lower signal frequencies
(D) higher signal frequencies

Open solution

Q23. GATE ECE 1998

For a given data rate, the bandwidth B_p of a BPSK signal and the band width B_0 of the OOK signal are related as

(A) $B_p = B_0/4$ (B) $B_p = B_0/2$ (C) $B_p = B_0$ (D) $B_p = 2B_0$

Open solution

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Q24. GATE ECE 1997

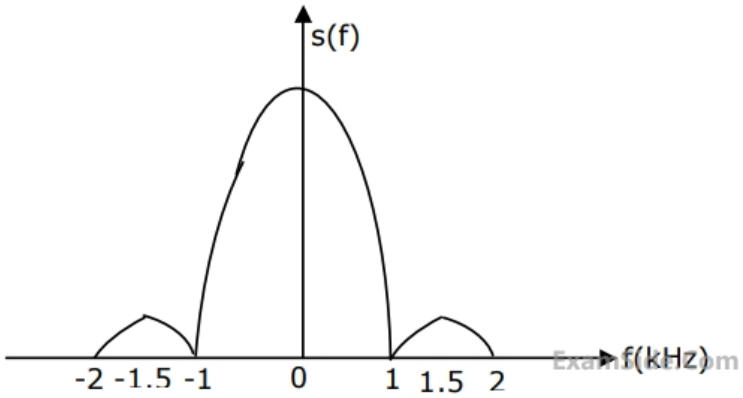
The line code that has zero dc component for pulse transmission of random binary data is

(A) Non - return to zero (NRZ)
(B) Return to zero(RZ)
(C) Alternate Mark Inversion (AMI)
(D) None of the above.

Open solution

Q25. GATE ECE 1997

A deterministic signal has the power spectrum given in figure. The minimum sampling rate needed to completely represent this signal is



(A) 1 KHz
(B) 2 KHz
(C) 3 KHz
(D) None of the above

Open solution

Q26. GATE ECE 1995

If the number of bits per sample in a PCM system is increased from n to n+1, the improvement in signal-to-quantization noise ratio will be

(A) 3 dB
(B) 6 dB
(C) 2n dB
(D) n dB

Open solution

Q27. GATE ECE 1995

The signal to quantization noise ratio in an n-bit PCM system

(A) depends upon the sampling frequency employed.
(B) is independent of the value of 'n'.
(C) increasing with increasing value of 'n'.
(D) decreases with the increasing value of n'.

Open solution

Q28. GATE ECE 1994

The bandwidth required for the transmission of a PCM signal increases by a factor of _____ when the number of quantization levels is increased from 4 to 64.

Open solution

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Marks 2 39 questions

Q29. GATE ECE 2026

The average bit error rate at the input of a $(7, 4, 1)$ Hamming decoder is 0.10 . The probability that the decoder will fail to decode a received word correctly is ____ . (rounded off to two decimal places)

Open solution

Q30. GATE ECE 2024

The information bit sequence {1 1 1 0 1 0 1 0 1} is to be transmitted by encoding with Cyclic Redundancy Check 4 (CRC-4) code, for which the generator polynomial is $C(x) = x^4 + x + 1$. The encoded sequence of bits is ____.

Open solution

(A) {1 1 1 0 1 0 1 0 1 1 1 0 0}
(B) {1 1 1 0 1 0 1 0 1 1 1 0 1}
(C) {1 1 1 0 1 0 1 0 1 1 1 1 0}
(D) {1 1 1 0 1 0 1 0 1 0 1 0 0}

Q31. GATE ECE 2022

Consider a channel over which either symbol x_A or symbol x_B is transmitted. Let the output of the channel Y be the input to a maximum likelihood (ML) detector at the receiver. The conditional probability density functions for y given x_A and x_B are :

$f_{Y|x_A}(y) = e^{-(y+1)}u(y+1),$

$f_{Y|x_B}(y) = e^{(y-1)}(1 - u(y-1)),$

where, u(.) is the standard unit step function. The probability of symbol error for this system is _____ (rounded off to two decimal places).

Open solution

Q32. GATE ECE 2020

In a digital communication system, a symbol S randomly chosen from the set $\{s_1, s_2, s_3, s_4\}$ is transmitted. It is given that $s_1 = -3, s_2 = -1, s_3 = +1$ and $s_4 = +2$. The received symbol is $Y = S + W$. W is a zero mean unit - variance Gaussian random variable and is independent of S. P_i is the conditional probability of symbol error for the maximum likelihood (ML) decoding when the transmitted symbol $S = s_i$. The index i for which the conditional symbol error probability P_i is the highest is ____ .

Open solution

Q33. GATE ECE 2017 Set 1

In binary frequency shift keying (FSK), the given signal waveforms are
$u_o(t) = 5\ \cos(20000\,\pi\,t);\ 0 \leq\ t\ \leq T$, and
$u_1(t) = 5\ \cos(22000\,\pi\,t);\ 0 \leq\ t\ \leq T$,
where T is the bit-duration interval and t is in seconds. Both $u_o(t)$ and $u_1(t)$ are zero outside the interval $0 \leq\ t\ \leq T$. With a matched filter (correlator) based receiver, the smallest positive value of T (in milliseconds) required to have $u_o(t)$ and $u_1(t)$ uncorrelated is

Open solution

(A) 0.25 ms (B) 0.5 ms (C) 0.75 ms (D) 1.0 ms

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Q34. GATE ECE 2016 Set 2

An ideal band-pass channel 500Hz-2000 Hz is deployed for communication. A modem is designed to transmit bits at the rate of 4800 bits/s using 16-QAM. The roll-off factor of a pulse with a raised cosine spectrum that utilizes the entire frequency band is_____

Open solution

Q35. GATE ECE 2016 Set 3

The bit error probability of a memoryless binary symmetric channel is 10^{-5}. If 10^{-5} bits are sent over this channel, then the probability that not more than one bit will be in error is _____

Open solution

Q36. GATE ECE 2015 Set 1

A sinusoidal signal of 2 kHz frequency is applied to a delta modulator. The sampling rate and step-size Δ of the delta modulator are 20,000 samples per second and 0.1 V, respectively. To prevent slope overload, the maximum amplitude of the sinusoidal signal (in Volts) is

(A) $\frac{1}{2\pi}$ (B) $\frac{1}{\pi}$ (C) $\frac{2}{\pi}$ (D) π

Open solution

Q37. GATE ECE 2015 Set 1

The transmitted signal in a GSM system is of 200 kHz bandwidth and 8 users share a common bandwidth using TDMA. If at a given time 12 users are talking in a cell, the total bandwidth of the signal received by the base station of the cell will be at least (in kHz) _____.

Open solution

Q38. GATE ECE 2014 Set 4

An M - level PSK modulation scheme is used to transmit independent binary digits over a band-pass channel with bandwidth 100 kHz. The bit rate is 200 kbps and the system characteristic is a raised-cosine spectrum with 100% excess bandwidth. The minimum value of M is _____.

Open solution

Q39. GATE ECE 2014 Set 2

The input to a 1-bit quantizer is a random variable X with pdf $f_x(x) = 2e^{-2x}\ for\ x \geq 0$ and $f_x(x) = 0\ for\ x < 0$. For outputs to be of equal probability, the quantizer threshold should be _____.

Open solution

Q40. GATE ECE 2014 Set 3

In a PCA system, the signal $m(t) = \{\sin(100\,\pi\,t) + \cos(100\,\pi\,t\}$ V is sampled at the Nyquist rate. The samples are processed by a uniform quantizer with step size 0.75 V. The minimum data rate of the PCM system in bits per second is_____

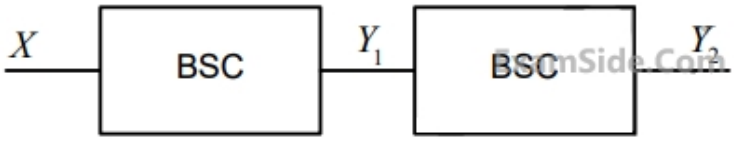
Open solution

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Q41. GATE ECE 2014 Set 3

A binary random variable X takes the value of 1 with probability 1/3. X is input to a casade of 2 independent identical binary symmetric channels (BSCs) each with crossover probability 1/2. The output of BSCs are the random variables Y_1 and Y_2 as shown in the figure.

Open solution



Q42. GATE ECE 2008

Four messages band limited to W, W, 2 W and 3 W respectively are to be multiplexed using Time Division Multiplexing (TDM). The minimum bandwidth required for transmission of this TDM signal is

Open solution

(A) 1 W (B) 3 W (C) 6 W (D) 7 W

Q43. GATE ECE 2008

A speech signal, band limited to 4 kHz and peak voltage varying between + 5V and - 5V is sampled at the Nyquist rate. Each sample is quantized and represented by 8 bits.
Assuming the signal to be uniformly distributed between its peak values, the signal to noise ratio at the quantizer output is

Open solution

(A) 16 dB (B) 32 dB (C) 48 dB (D) 64 dB

Q44. GATE ECE 2008

A speech signal, band limited to 4 kHz and peak voltage varying between + 5V and - 5V is sampled at the Nyquist rate. Each sample is quantized and represented by 8 bits.
If the bits 0 and 1 are transmitted using bipolar pulses, the minimum bandwidth required for distortion free transmission is

Open solution

(A) 64 kHz (B) 32 kHz (C) 8 kHz (D) 4 kHz.

Q45. GATE ECE 2008

A speech signal, band limited to 4kHz and peak voltage varying between +5V and -5V is sampled at the Nyquist rate. Each sample is quantized and represented by 8 bits.
The number of quantitization levels required to reduce the quantization noise by a factor of 4 would be

Open solution

(A) 1024 (B) 512 (C) 256 (D) 64

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Q46. GATE ECE 2007

In delta modulation, the slope overload distortion can be reduced by

(A) decreasing the step size
(B) decreasing the granular noise
(C) decreasing the sampling rate
(D) increasing the step size

Open solution

Q47. GATE ECE 2007

In a GSM system, 8 channels can co-exist in 200 KHz bandwidth using TDMA. A GSM based cellular operator is allocated 5 MHz bandwidth. Assuming a frequency reuse factor of 1/5, i. e. a five-cell repeat pattern,, the maximum number of simultaneous channels that can exist in one cell is

(A) 200 (B) 40 (C) 25 (D) 5

Open solution

Q48. GATE ECE 2007

The raised cosine pulse p(t) is used for zero ISI in digital communications. The expression for p(t) with unity roll-off factor is given by

$$p(t) = \frac{\sin 4\pi W t}{4\pi W t(1 - 16W^2t^2)}$$

The value of p(t) at $t = \frac{1}{4W}$ is

(A) - 0.5 (B) 0 (C) 0.5 (D) ∞

Open solution

Q49. GATE ECE 2007

In a Direct sequence CDMA system the chip rate is 1.2288×10^6 chips per second. If the processing gain is desired to be at least 100. The data rate

(A) must be less than or equal to 12.288×10^3 bits per sec.
(B) must be greater than 12.288×10^3 bits per sec.
(C) must be exactly equal to 12.288×10^3 bits per sec.
(D) can take any value less than 12.288×10^3 bits per sec.

Open solution

Q50. GATE ECE 2006

The minimum step-size required for a Delta-Modulator operating at 32 K samples/sec to track the signal (here u(t) is the unit-step function)
x(t) = 125t(u(t) - u (t - 1) + (250 - 125t) (u (t - 1) - u (t - 2)) so that slope - overload is avoided, would be

(A) 2^{-10} (B) 2^{-8} (C) 2^{-6} (D) 2^{-4}

Open solution

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Q51. GATE ECE 2004

Three analog signals, having bandwidths 1200 Hz, 600 Hz and 600 Hz are sampled at their respective Nyquist rates, enocodes with 12 bit words, and time division multiplexed. The bit rate for the multiplexed signal is

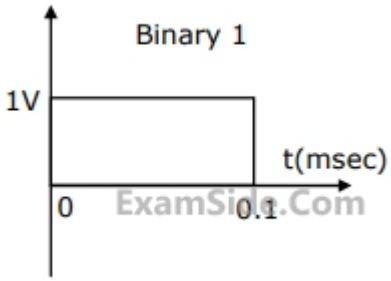
Open solution

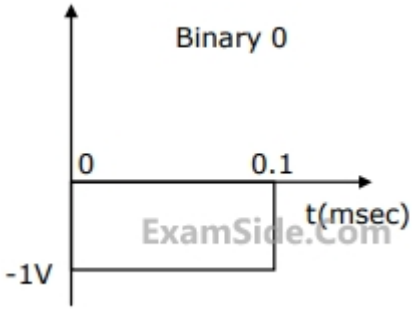
(A) 115.2 kbps (B) 28.8 kbps (C) 57.6 kbps (D) 38.4 kbps

Q52. GATE ECE 2004

A source produces binary data at the rate of 10 kbps. The binary symbols are represented as shown in fig.

Open solution





The source output is transmitted using two modulation schemes, namely Binary PSK (BPSK) and Quadrature PSK (QPSK). Let B_1 and B_2 be the bandwidth requirements of BPSK and QPSK respectively. Assuming that the bandwidth of the above rectangular pulses is 10 kHz, B_1 and B_2 are

(A) B_1 = 20 KHz, B_2 = 20 KHz (B) B_1 = 10 KHz, B_2 = 20 KHz

(C) B_1 = 20 KHz, B_2 = 10 KHz (D) B_1 = 10 KHz, B_2 = 10 KHz

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Q53. GATE ECE 2004

Open solution

Choose the correct one from among the alternatives A, B, C, D after matching an item from Group 1 with the most appropriate item in Group 2.

Group

1. FM
2. DM
3. PSK
4. PCM

Group 2

P: Slope overload
Q: $\mu - law$
R: Envelope detector
S: Capture effect
T: Hilbert transform
U: Matched filter

(A) 1 - T, 2 - P, 3 - U, 4 - S
(B) 1 - S, 2 - U, 3 - P, 4 - T
(C) 1 - S, 2 - P, 3 - U, 4 - Q
(D) 1 - U, 2 - R, 3 - S, 4 - Q

Q54. GATE ECE 2003

Open solution

The input to a linear delta modulator having a step size Δ = 0.628 is a sine wave with frequency f_m and peak amplitude E_m. If the sampling frequency f_s = 40 KHz. the combination of the sine-wave frequency and the amplitude, where slope overload will take place is

(A) Em 0.3V, Fm 8 KHz
(B) Em 1.5V, Fm 4 KHz
(C) Em 1.5V, Fm 2 KHz
(D) Em 3.0V, Fm 1 KHz

Q55. GATE ECE 2003

Open solution

If S represents the carrier synchronization at the receiver and ρ represents the bandwidth efficiency, then the correct statement for the coherent binary PSK is

(A) $\rho = 0.5,\ S$ is required
(B) $\rho = 1.0,\ S$ is required
(C) $\rho = 0.5,\ S$ is not required
(D) $\rho = 1.0,\ S$ is not required

Q56. GATE ECE 2003

Open solution

A signal is sampled at 8 KHz and is quantized using 8-bit uniform quantizer. Assuming SNR_q for a sinusoidal signal, the correct statement for PCM signal with a bit rate of R is

(A) R = 32 kbps, SNR_q = 25.8 dB
(B) R = 64 kbps, SNR_q = 49.8 dB
(C) R = 64 kbps, SNR_q = 55.8 dB
(D) R = 32 kbps, SNR_q = 49.8 dB

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Q57. GATE ECE 2000

In a digital communication system employing Frequency Shift Keying (FSK), the 0 and 1 bit are represented by sine waves of 10 KHz and 25 KHz respectively. These waveforms will be orthogonal for a bit interval of

(A) $45\ \mu\sec$ (B) $200\ \mu\sec$ (C) $50\ \mu\sec$ (D) $250\ \mu\sec$

Open solution

Q58. GATE ECE 1999

Four independent messages have bandwidths of 100Hz, 100 Hz, 200Hz and 400Hz respectively. Each is sampled at the Nyquist rate, and the samples are time division multiplexed (TDM) and transmitted. The transmitted sample rate)in Hz) is

(A) 1600 (B) 800 (C) 400 (D) 200

Open solution

Q59. GATE ECE 1996

The number of bits in a binary PCM system is increased from n to n+1. As a result, the signal to quantization noise ratio will improve by a factor

(A) $(n+1)/n$ (B) $2^{(n+1)/n}$

(C) $2^{2(n+1)/n}$ (D) which is independent of n

Open solution

Q60. GATE ECE 1992

The bit stream 01001 is differentially encoded using 'Delay and Ex OR' scheme for DPSK transmission. Assuming the reference bit as a '1' and assigning phases of '0' and 'π' for 1's and 0's respectively, in the encoded sequence, the transmitted phase sequence becomes

(A) $\pi\ 0\ \pi\ \pi\ 0$ (B) $0\ \pi\ \pi\ 0\ 0$ (C) $0\ \pi\ \pi\ \pi\ 0$ (D) $\pi\ \pi\ 0\ \pi\ \pi$

Open solution

Q61. GATE ECE 1992

Coherent demodulation of FSK signal can be effected using

(A) correlation receiver (B) band pass filters and envelope detectors

(C) matched filter (D) discriminator detection

Open solution

Q62. GATE ECE 1992

Source encoding in a data communication system is done in order to

(A) enhance the information transmission.

(B) band pass filters and envelope rate detectors.

(C) conserve the transmitted power.

(D) discriminator detection.

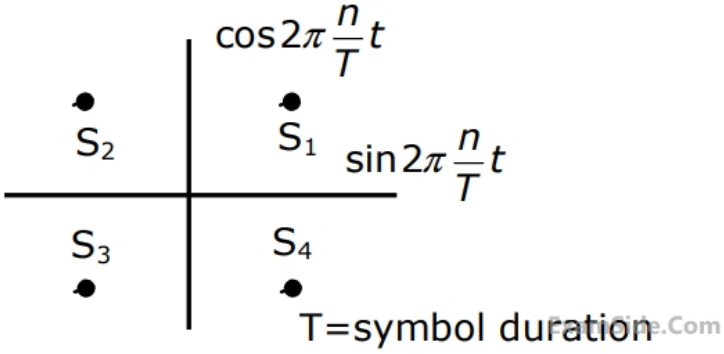
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Q63. GATE ECE 1991

For the signal constellation shown in the figure, the type of modulation is _____.

Open solution



Q64. GATE ECE 1990

In a BPSK signal detector, the local oscillator has a fixed phase error of 20^0. This phase error deteriorates the **SNR** at the output by a factor of

(A) $\cos 20^0$ (B) $\cos^2 20^0$ (C) $\cos 70^0$ (D) $\cos^2 70^0$

Open solution

Q65. GATE ECE 1989

In binary data transmission DPSK is preferred to PSK because

(A) a coherent carrier is not required to be generated at the receiver.

(B) for a given energy per bit, the probability of error is less.

(C) the 180^0 phase shifts of the carrier are unimportant.

(D) more protection is provided against impulse noise.

Open solution

Q66. GATE ECE 1988

The transfer function of a zero - order hold is

(A) $\frac{1-\exp(-Ts)}{s}$ (B) $\frac{1-\exp(-Ts)}{s}$

(C) 1 (D) $\frac{1}{[1-\exp(-Ts)]}$

Open solution

Q67. GATE ECE 1988

The message bit sequence input to a DPSK modulator is 1, 1, 0, 0, 1, 1. The carrier phase during the reception of the first two message bits is π, π. The carrier phase for the remaining four message bits is

(A) $\pi, \pi, 0, \pi$ (B) $0, 0, \pi, \pi$ (C) $0, \pi, \pi, \pi$ (D) $\pi, \pi, 0, 0$

Open solution

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Marks 8 1 questions

Q68. GATE ECE 1987

A rectangular pulse existing between t = 0 and t = T and having an amplitude A is to be received by a matched filter
(a) Draw the impulse response of the matched filter labeling all important points and the axes.
(b) Find the signal shape at the filter output. Lable all important points and axes.

Open solution

Marks 10 2 questions

Q69. GATE ECE 1989

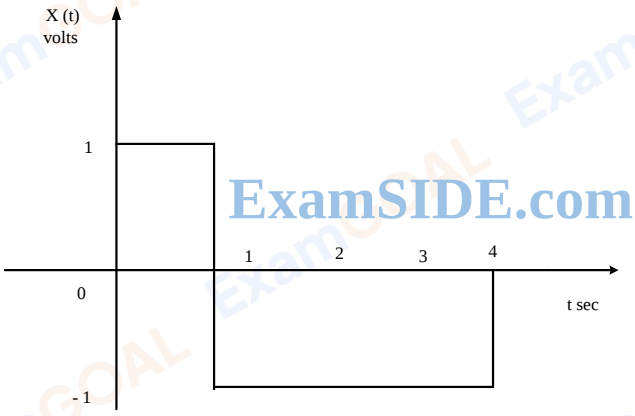
A signal of bandwidth 3.5 kHz has peak-to-peak and rms amplitudes of 4 V and 0.2 V, respectively. This signal is passed through a linear PCM coder and then transmitted over a channel capable of supporting a transmission rate of 60 kbps. Calculate the maximum signal-to-quantization-noise ratio that can be obtained using this system.

Open solution

Q70. GATE ECE 1989

For the input signal, x(t), shown in Fig.
(a) Sketch the impulse response, h(t), of the matched filter.
(b) Sketch the output, y(t), of the matched filter. Fully label your diagram.
(c) If the input to the matched filter consists of the signal, x(t), plus additive white noise of two sided spectral density of 10^{-6} watts/Hz, find the maximum output SNR of the matched filter.

Open solution



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Answer Key

Marks 1

1. **C**	2. **(1) (0.95 to 1.05)**	3. **A**	4. **A, B, D**	5. **3**
6. **30.5 to 30.8**	7. **3.10 to 3.20**	8. **A**	9. **B**	10. **D**
11. **16**	12. **35**	13. **D**	14. **8**	15. **D**
16. **C**	17. **A**	18. **A**	19. **D**	20. **C**
21. **D**	22. **A**	23. **C**	24. **C**	25. **B**
26. **B**	27. **C**	28. **3**		

Marks 2

29. **(0.15) (0.10 to 0.20)**	30. **A**	31. **0.23**	32. **3**	33. **B**
34. **0.25**	35. **0.735**	36. **A**	37. **400**	38. **16**
39. **0.345**	40. **200**	41. **2**	42. **D**	43. **C**
44. **A**	45. **B**	46. **D**	47. **B**	48. **C**
49. **A**	50. **B**	51. **C**	52. **C**	53. **C**
54. **B**	55. **A**	56. **B**	57. **B**	58. **A**
59. **D**	60. **C**	61. **A**	62. **A**	63. **Quadrature phase shift keying**
64. **B**	65. **A, C**	66. **A**	67. **C**	

Marks 8

68. **solve it**	

Marks 10

69. **33.4 dB.**	70. **(c)** 4×10^6

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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